

LAYOUT

Title - Realization Design.

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Date - 17/05/2028

Sheet No - Sheet # 5

Description - Interactive narrative visualization communicating findings about educational infrastructure investment socioeconomic equity & household travel behaviour in Victoria to policy makers & transport planners.

OPERATIONS

- [DP] (Click "Jump to explore" - State A becomes State B. Right zone activates.
- [EB] Natural scroll past section break. None effect as clicking [DP]. Right zone activates.
- [A] Reader scrolls left zone ->
- Hover any dot in [P2] after -> Tooltip: LGA. Project count, household count, investment.
- Hover end point in [P3] slope -> Tooltip: T100, mode '1', n trips.
- Hover dot in [P4] connected dot -> Tooltip: ISRD ranking, mode '1', share, example LGA at the ranking.
- [H] (Click source link -> Open VISTA open data URL in new browser tab.
- [I] (Click < Back to narrative

Default on load: State A
(Right zone greyed)

Hover tooltip work in both states.
Only ops 1 & 2 change to state.

[A] Scroll

Click to explore
Hidden on DP

[ET]

Jump to Explore

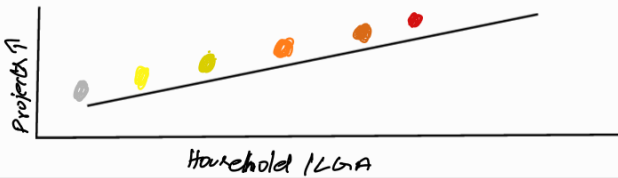
[DP]

< Back to narrative [ET]

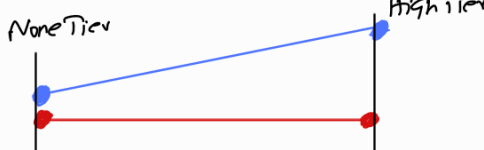
[P1] Victorian Education Investment & Travel Equity.

91 victoria investing in the right place? And does it matter how families travel? Scroll <

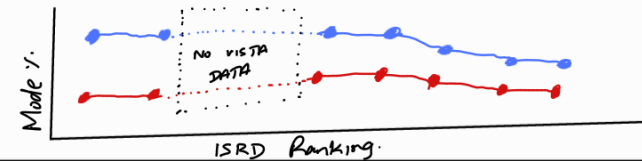
[P2] Headline: Investment broadly follows households



[P3] More Investment = More can use.



[P3] This is not about investment. It is about who can afford to live near transport option.



Now Explore the evidence yourself.

[EB] Section Break. Appears after P4.

FOCUS PART 1

State A - Guided Scroll

P1 Title

P2 Scatter

P3 Slope

P4 Connected Dot

- Hover tooltips only.
- Right zone greyed.
- [DP][EB] enable state B.
- P1 -> P2 -> P3 -> P4

State B - Supporting Evidence

< Back to narrative

[P2] Policy

Implication Text

[P3] 29 LGA

Gap

[H] Source

Link.

Narrative Block
P1-P4 are still
scrollable.

- Hover tooltips across all panels.
- Static text + callout - no interaction.
- [ET] to scroll to [P3] state removed.
- No filters. No dropdowns.

DETAILS Algorithm.

- reactive() + reactiveVal()
- qqplotly() wrapper
- geom_smooth
- geom_segment + geom_point()
- geom_point() + geom_line()
- annotate("text") + annotate("rule")
- scroll() -> JS -> shinyjs [A]

R Shiny Libraries.

- shiny 2.1.8.0 - base reactive framework
- qqplotly 3.5.0 - All charts
- plotly 4.10.0 - Tooltips.
- dplyr 1.1.0 -> Data Filtering
- survey 4.1.2.0 -> Survey weighted for vehicle ownership
- shinyjs 2.1.0 - AIB toggle.
- bslib 0.6.0 - Bootstrap.
- htmltools - Custom HTML
- scales, readr, here - Axis labels.
- Data files.

Time Estimates

- Scaffold + data loading - 4hr
- P1 - title & framing panel - 1hr
- P2 - scatter - 4hr
- P3 - slope - 4hr
- P4 - connected dot - 4hr
- [A] - scroll bar - 1hr
- Right zone - 4hr
- AIB toggle - 4hr.
- Styling - 2hr
- Testing & Refug - 4hr

Total - 32hrs.

Hardware / Notes.

- Browser - Chrome 120+, Firefox 120+, Safari 17+.
- Deployment:
- Dataset < 50MB.
- No GPU.
- Narrative + DEP distinction.
- Each panel: Headline -> (chart -> callout -> chart prove stated claims. Reader does not interpret. DEP asks questions DUP (xfer) answers. User charts as evidence.
- P1 asks -> P2 where P3 -> Now what -> P4 why.

Lineage.

- 1) Martini glass Architecture (Sheet 4)
- 2) [A] Natural bar concept (Sheet 2)
- 3) [P3] Slope chart (Sheet 1)
- 4) Right zone filter removed -> Policy Text replaced (Sheet 4)
- 5) P2 scatter + P4 connected (Sheet 3 [ET] & [DP])

